BLOCKCHAIN LAB (Task:01)

1. What is a block in blockchain, and what are its main components?

A **block** is a collection of transactions/data that is added to the blockchain. Each block is linked to the previous block using its hash.

Main components:

- **Block Header** – contains metadata such as previous block hash, timestamp, and nonce.
- **Transactions/Data** – records of transactions stored in the block.
- **Hash** – unique value generated from the block's contents.
- **Previous Block Hash** – connects the current block to the previous block.

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2. What is a nonce, and why do miners use it?

A **nonce (number used once)** is a value that miners change repeatedly while trying to find a valid block hash.

In **Proof of Work (PoW)**, miners change the nonce until the resulting hash satisfies the network's difficulty requirement.

Example:
```
Data + Nonce → SHA-256 → Hash
```

☞ **Purpose:** To find a valid hash and prove that computational work was performed.

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3. What is a hash function, and why is it used in blockchain?

A **hash function** converts input data of any size into a fixed-length, unique-looking value called a **hash**.

Blockchain uses hashing to:

- Maintain **data integrity**
- Link blocks together
- Detect unauthorized changes
- Support mining/Proof of Work

Example:
```
Transaction Data → Hash Function → Hash
```

Even a small change in the input produces a very different hash.

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4. What is SHA-256, and how is it used in blockchain technology?

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hash function that produces a **256-bit (64 hexadecimal character)** hash.

It is used in systems such as **Bitcoin** for:

- Generating block hashes
- Linking blocks
- Supporting Proof of Work
- Maintaining data integrity

☞ **Key point:** SHA-256 produces a fixed-size **256-bit hash**.

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5. What are the four main parts of blockchain transactions?

A transaction commonly contains:

1. **Sender/Input** – identifies the source of the funds/data.
2. **Receiver/Output** – identifies the destination.
3. **Amount/Data** – specifies what is being transferred or recorded.
4. **Digital Signature** – proves that the transaction was authorized by the sender.

☞ In a cryptocurrency context, the exact transaction structure can vary between blockchains.

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6. What are public and private keys, and what is their purpose in blockchain?

Public key: A key that can be shared with others and is used to identify an account or verify digital signatures.

Private key: A secret key that must be kept secure. It is used to **create/sign transactions**.

Purpose:

- Public key → identification/verification
- Private key → authorization/signing

☞ **Never share a private key.**

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7. What are the major advantages and disadvantages of blockchain technology?

Advantages:

- **Decentralization** – no single central authority controls the network.
- **Transparency** – transactions can be verified by network participants.
- **Immutability** – recorded data is difficult to alter.
- **Security** – cryptography protects transactions and data.
- **Traceability** – transactions can be tracked through the ledger.

Disadvantages:

- High energy consumption in some PoW blockchains.
- Scalability problems.
- Transaction speed can be slower than centralized systems.
- Storage requirements can grow significantly.
- Some blockchain systems have high transaction fees.

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8. How does blockchain maintain security and protect data from unauthorized modification? (3 marks)

Blockchain protects data mainly through **hashing, cryptography, and decentralization**.

1. Hashing:
Each block has a hash based on its contents. If someone changes the data, the hash changes, making the modification detectable.

2. Previous Hash:
Each block stores the hash of the previous block. Therefore, changing one block breaks the chain of hash references.

3. Consensus + Cryptography:
Consensus mechanisms such as **Proof of Work** help prevent unauthorized changes, while **digital signatures** ensure that only the holder of the appropriate private key can authorize a transaction.

⋆ 3-mark exam version

Blockchain maintains security using hashing, cryptography, and consensus mechanisms. Each block contains its own hash and the previous block's hash, so changing data makes the hash invalid and breaks the chain. Digital signatures verify transaction ownership, while consensus mechanisms prevent unauthorized changes from being accepted by the network.

Blockchain Lab — 20 Short Questions with Answers

9. What is a blockchain ledger?

A **blockchain ledger** is a distributed digital record that stores transactions or data in a sequence of blocks. It is shared among network participants and is difficult to modify.

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10. What is Proof of Work (PoW)?

Proof of Work is a consensus mechanism where miners solve a computational problem to add a new block to the blockchain.

☞ **Purpose:** To verify blocks and prevent fraudulent activities.

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11. What is Proof of Stake (PoS)?

Proof of Stake is a consensus mechanism where validators are selected based partly on the amount of cryptocurrency they **stake**.

☞ It generally requires much less energy than Proof of Work.

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12. What is a Genesis Block?

The **Genesis Block** is the **first block of a blockchain**.

It does not point to a previous blockchain block and is used as the starting point of the chain.

☞ **Example:** Bitcoin's first block is called the Genesis Block.

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13. What is a Merkle Tree?

A **Merkle Tree** is a tree-like structure that organizes transaction hashes into a single value called the **Merkle Root**.

☞ It helps efficiently verify whether a transaction is included in a block.

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14. What is a Digital Signature in blockchain?

A **digital signature** is a cryptographic method used to prove that a transaction was authorized by the owner of a private key.

- **Private key → signs**

- **Public key → verifies**

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15. What is cryptocurrency?

Cryptocurrency is a digital or virtual currency that uses cryptography for security and typically operates on a blockchain or similar distributed ledger.

☞ **Examples:** Bitcoin and Ether.

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16. What is decentralization in blockchain?

Decentralization means that control and data are distributed among multiple computers or nodes instead of being controlled by a single central authority.

☞ Example: Bitcoin does not depend on one central bank to maintain its ledger.

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17. What is consensus in blockchain?

Consensus is the process through which blockchain network participants agree on the valid state of the blockchain.

☞ Examples:

- Proof of Work
- Proof of Stake

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18. What is a blockchain node?

A **node** is a computer connected to the blockchain network that participates in storing, transmitting, validating, or processing blockchain data.

☞ Multiple nodes help maintain the distributed nature of the blockchain.

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19. What is a Peer-to-Peer (P2P) network?

A **P2P network** is a network where computers communicate directly with each other without requiring a central server.

Blockchain networks commonly use P2P communication to share transactions and blocks.

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20. What is double spending?

Double spending is the attempt to spend the **same digital currency more than once**.

Blockchain consensus mechanisms help prevent the same cryptocurrency from being successfully used in conflicting transactions.

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21. What is a smart contract?

A **smart contract** is a program stored on a blockchain that automatically executes predefined rules when specified conditions are met.

☞ **Example:** Automatically transferring digital assets when a payment condition is satisfied.

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22. What is Ethereum?

Ethereum is a blockchain platform that supports **smart contracts and decentralized applications (dApps)**.

Its native cryptocurrency is called **Ether (ETH)**.

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23. What is Bitcoin?

Bitcoin is a decentralized digital cryptocurrency that operates on a blockchain.

It uses **Proof of Work** as its consensus mechanism and was introduced by the pseudonymous **Satoshi Nakamoto**.

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24. What is a 51% attack?

A **51% attack** occurs when an entity or group controls a majority of the network's relevant consensus power, such as hash power in a PoW blockchain.

This can allow attackers to potentially **reorganize recent transactions or perform double-spending attacks**.

☞ It does not automatically mean they can steal everyone's funds or change arbitrary old data.

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25. What is a fork in blockchain?

A **fork** occurs when the blockchain's rules or software are changed, potentially causing the chain to split into different versions.

Two common types are:

- **Soft fork** → backward-compatible rule change
- **Hard fork** → non-backward-compatible rule change

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26. What is the difference between public and private blockchain?

Public Blockchain	Private Blockchain
Open to anyone	Restricted participants
Usually decentralized	Usually controlled by an organization
Example: Bitcoin	Used in enterprise systems

☞ **Simple:** Public = open participation; Private = permissioned participation.

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27. What is a Block Header?

A **block header** contains important metadata about a block.

Common fields include:

- Previous block hash
- Timestamp
- Nonce
- Merkle Root
- Difficulty/target information in PoW systems

☞ It helps identify and validate the block.

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28. What is the role of timestamps in blockchain?

A **timestamp** records the approximate time associated with a block.

It helps:

- Maintain the chronological order of blocks
- Provide information about when a block was created
- Support validation and network operation

🔥 Extra 10 Very Likely Questions

If your teacher wants slightly more **lab-oriented** questions, also prepare these:

29. What is a block hash?

A **block hash** is a fixed-length cryptographic value generated from the block's data. It acts like a digital fingerprint of the block.

30. What happens if data inside a block is changed?

Changing the data changes the block's hash. This can make the block invalid and break its connection with subsequent blocks.

31. What is mining?

Mining is the process used by PoW blockchains where miners perform computational work to find a valid block and add it to the blockchain.

32. What is a validator?

A **validator** is a participant that helps verify transactions and blocks and maintain consensus, particularly in Proof of Stake systems.

33. What is a transaction hash?

A **transaction hash** is a cryptographic identifier generated from transaction data. It can be used to identify and verify a transaction.

34. What is immutability in blockchain?

Immutability means that once data is recorded and sufficiently confirmed on a blockchain, changing it is difficult because modifications can be detected and may require overcoming the network's consensus/security mechanisms.

35. What is transparency in blockchain?

Transparency means that on many public blockchains, transaction records can be viewed and independently verified by network participants or the public.

36. What is a distributed ledger?

A **distributed ledger** is a database whose synchronized copies are maintained across multiple network participants rather than one central location.

37. What is a blockchain explorer?

A **blockchain explorer** is a tool or website that allows users to view blockchain information such as blocks, transactions, and addresses.

38. What is the difference between blockchain and a traditional database?

A **traditional database** is usually controlled by an administrator or organization, while a **blockchain** distributes record-keeping and uses cryptographic linking and consensus mechanisms to maintain the ledger.

⭐ Last-Minute Priority

For tomorrow's exam, I would memorize these **15 first**:

Block → Hash → SHA-256 → Nonce → Mining → PoW → PoS → Public/Private Key → Digital Signature → Genesis Block → Merkle Tree → Consensus → Double Spending → Smart Contract → 51% Attack